Group Topic: EduPulse | Lab No. WEEK #2

Git Repo/ Colab Link: https://github.com/Drexgithub/Group-Repository | Date: 04/23/2026

Objectives

The objective of this task was to procure and process a large-scale dataset (minimum 500,000 rows) from a single source while adhering to strict data ethics and procurement standards. Post-acquisition, the workflow focused on rigorous data cleaning and distributed processing to ensure the dataset was optimized for high-fidelity analysis.

Tasks Completed

1. Data Procurement & Ethical Scrubbing

- **Sourcing:** Acquired a **Student Performance** dataset containing 1,000,000 records from a structured CSV.
- **Downsampling:** Trimmed the dataset to exactly **500,000 rows** to meet specific task requirements.
- **Ethical Compliance:** Identified and removed sensitive columns—attendance_percentage and class_participation—to prevent proxy bias and maintain privacy standards.
- **Output:** Saved the refined data as student_performance_clean.csv, featuring four core attributes: student_id, weekly_self_study_hours, total_score, and grade.

2. Load Dataset into a Spark DataFrame

The cleaned dataset was ingested into a Spark DataFrame to leverage distributed computing. The schema was automatically inferred, and data was partitioned across executors to ensure scalability.

```
from pyspark.sql import SparkSession # Initialize Spark Session

        spark = SparkSession.builder.appName('StudentPerformance').getOrCreate()

        df = spark.read.csv('student_performance_clean.csv', header=True,

inferSchema=True)

df.printSchema()

df.show(5)
```

3. Exploratory Data Analysis (EDA)

```
>>> # Check missing/null values per column
>>> print("=== Missing Values ===")
=== Missing Values ===
>>> df.select([
...     F.count(F.when(F.col(c).isNull(), c)).alias(c)
...     for c in df.columns
... ]).show()
+----------+----------------------+-----------+-----+
|student_id|weekly_self_study_hours|total_score|grade|
+----------+----------------------+-----------+-----+
|         0|                     0|          0|    0|
+----------+----------------------+-----------+-----+
```

EDA was conducted to validate data integrity and structural consistency.

Missing Values

The audit confirmed zero null values across all 500,000 records. The dataset is complete and requires no imputation.

4. Filtering Invalid Records

Filtering rules were applied to remove logically invalid records from the dataset:

- **Score Integrity:** Removed records where total_score > 100.
- **Temporal Logic:** Removed records where weekly_self_study_hours < 0.

Result: All 500,000 records passed validation, confirming a logically consistent dataset.

```
df_filtered = df.filter((df.total_score <= 100) & (df.weekly_self_study_hours >= 0))

print(f'Records after filtering: {df_filtered.count()}')
```

Tasks to be Completed

- Create data visualizations (histograms, scatter plots, box plots) using matplotlib/seaborn to illustrate grade distribution and outlier spread.
- Perform time-series analysis if timestamped data becomes available in future dataset iterations.

Results and Analysis

Grade Distribution

```
>>>
>>> grade_dist.show()
+-----+------+----------+
|grade| count|percentage|
+-----+------+----------+
|    A|274151|     54.83|
|    B|129399|     25.88|
|    C| 70731|     14.15|
|    D| 22571|      4.51|
|    F|  3148|      0.63|
+-----+------+----------+
```

The population is characterized by high academic achievement, with 54.83% of students receiving an "A" grade. The distribution exhibits a right-skewed pattern, indicating a concentration of high performers with a tapering frequency in lower grade brackets.

Data Quality

student_id	weekly_self_study_hours	total_score	grade
0	0	0	0

The dataset achieved a 100% integrity score (zero missing or invalid records post-filtering). This high level of reliability provides a robust foundation for future predictive modeling or machine learning applications.

Challenges and Solutions

Challenge: The original dataset contained 1,000,000 rows, exceeding the 500,000-row requirement.

Solution: Used pandas head(500000) to trim the dataset to exactly 500,000 rows before loading into Spark.

Challenge: Dataset contained ethically sensitive proxy variables (attendance and participation) that could introduce bias.

Solution: Dropped attendance_percentage and class_participation columns during pre-processing to comply with data ethics standards.

Challenge: Memory constraints when loading the full 1M-row dataset for initial inspection.

Solution: Used Spark's distributed Data Frame architecture and pandas rows parameter for efficient sampling before full load.

Conclusion

This project successfully demonstrated a complete big data lifecycle—from ethical procurement to distributed processing. By utilizing Apache Spark, we maintained computational efficiency across a large volume of data. The final analysis reveals a significant correlation between effort and outcome, with A-grade students averaging over 13x more study time than those in the F-grade bracket, highlighting study hours as a critical predictor of success.